# FTI-1000 DC & AC CALIBRATION MANUAL

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# 1.0 INTRODUCTION

This manual describes the philosophy and procedures for calibrating/verifying FTI testers. The intended audience is calibration technicians and manufacturing engineers.

# 2.0 PHILOSOPHY

## 2.1 CALIBRATION PHILOSOPHY

Calibration of analog instrumentation improves the accuracy by measuring actual un-calibrated performance and applying correction during use. Calibration assumes instruments have linear transfer functions and corrects for offset and gain. The standard steps are as follows:

1. Measure two points on the transfer curve
2. Calculate offset and gain
3. Calculate calibration constants
4. Store calibration constants in the instrument

Post calibration, when the instrument is programmed, ideal values are sent to the instrument, and the instrument applies the calibration constants to correct for non-ideal hardware.

## 2.2 VERIFICATION PHILOSOPHY

Instruments are calibrated at two points and therefore there is no correction for non-linearity. Linearity is a by-product of instrument design. Verification is a process that tests the validity of the calibration and the linearity of the instrument. Measuring three points of the transfer curve to ensure the transfer curve is not bowed guarantees linearity.

## 2.3 NIST TRACEABILITY AND TRANSFER STANDARDS

Calibration must be traceable to NIST standards or an intermediate standards organization. Traceability of FTI calibration is via an Agilent 34401A Meter. The Meter can measure voltage and resistance and carries a certification of calibration and traceability from a calibration lab chosen by customers.

Not all calibration measurements are voltage, so a calibration box is used to create transfer standards. The calibration box contains precision resisters that are measured by the Meter at calibration time. Resisters are used as loads that convert output current of instruments to voltage. The Meter measures voltage and current and calculates using the Meter's voltage measurement and the resistance of the transfer standard.

Transfer standards are measured when calibration begins to remove any effects of temperature change.

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## 2.4 Automated Calibration and Verification

Manual calibration involves a written procedure whereby a technician attaches standards and external instruments to an instrument under calibration and makes measurements. Results are written down on paper and entered into the instrument under calibration via keyboard.

Automated calibration removes the need for manual connections other than an initial connection to a calibration box. In all FTI systems, calibration proceeds as follows:

1. Connect calibration box to system
2. Connect Meter to calibration box
3. Run calibration program from computer
4. Run verification program from computer

No data is entered manually and the Meter is connected to the tester via relays.

Automation results in the following benefits:

1. Prevents connection errors
2. Prevents measurement errors
3. Prevents data entry errors
4. Reduces cost through time savings
5. Allows FTI to update procedures through software releases

From a measurement point of view the connections made for manual and automated calibration would be the same. The main difference is the human element is removed so that the process takes less time and prevents human error. Even in the case where the technician misconnects the meter or calibration box is detected by software so that calibration is foolproof.

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# 3.0 System Set-up Requirements and Connections

This section will present the minimum set-up requirements and proper connections in performing system calibration.

## 3.1 Hardware Requirements

1. HP/Agilent 34401A Multi-meter (Calibration Standard)

   Calibration is performed with the HP/Agilent 34401A Multi-meter. FTI Studio can control this meter using the Serial port (RS-232). Make sure the settings on the Meter match the settings below: (Instructions for this are located in manual supplied by the manufacturer of meter)

   A. Set port to RS-232
   B. 9600 Baud
   C. No Parity
   D. 8 Bits
   E. 1 Start bit
   F. 2 Stop bits

2. Multi-meter interface cable (RS-232 to USB Cable) supplied by Focused Test, Inc.
3. FTI-1000 Calibration Box.
4. Various Banana Cables for making connections between the Calibration Box, Test Head & Meter
5. Ribbon Cable supplied by Focused Test, Inc. to connect Cal Box to Test Head.

## 3.2 Hardware Connections

Below is the guide for the proper hardware connections. Before performing this section ensure the system is in POWER DOWN or OFF.

### 3.2.1 Multi-meter Interface Cable (RS - 232 to USB) Connection

Since Calibration procedures require the use of HP/Agilent 34401A Multi-meter, we need to establish a working interface between the meter and the control computer.

Interface is established by the use of the RS -232 to USB cable connection. RS-232 to USB cable is expectedly provided by Focused Test Inc. However you can also make your own RS-232 to USB cable by following below instructions.

A. Create a DB-9 Null Modem Cable, Female to Female.
   Note: RS232 Port of HP/Agilent 34401A Multi-meter located at the back is in DB-9 Male Connector.
B. Purchase a USB - RS232 Converter Cable (DB-9 Male connector at one end then USB connector on the other end). USB – RS 232 Converter Cable normally comes with an installation CD, just install it to use cable.
C. Connect one female end of DB-9 Null Modem cable to RS 232 port of HP/Agilent 34401A Multi-meter.
D. Connect the other female end of DB-9 Null Modem cable to the DB-9 Male end of USB – RS 232 Converter Cable.
E. Plug in the USB connector end of USB – RS 232 Converter Cable to your system's USB port.

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F. Ready to use.
G. Refer to Figure 3.2.1A for the visual.

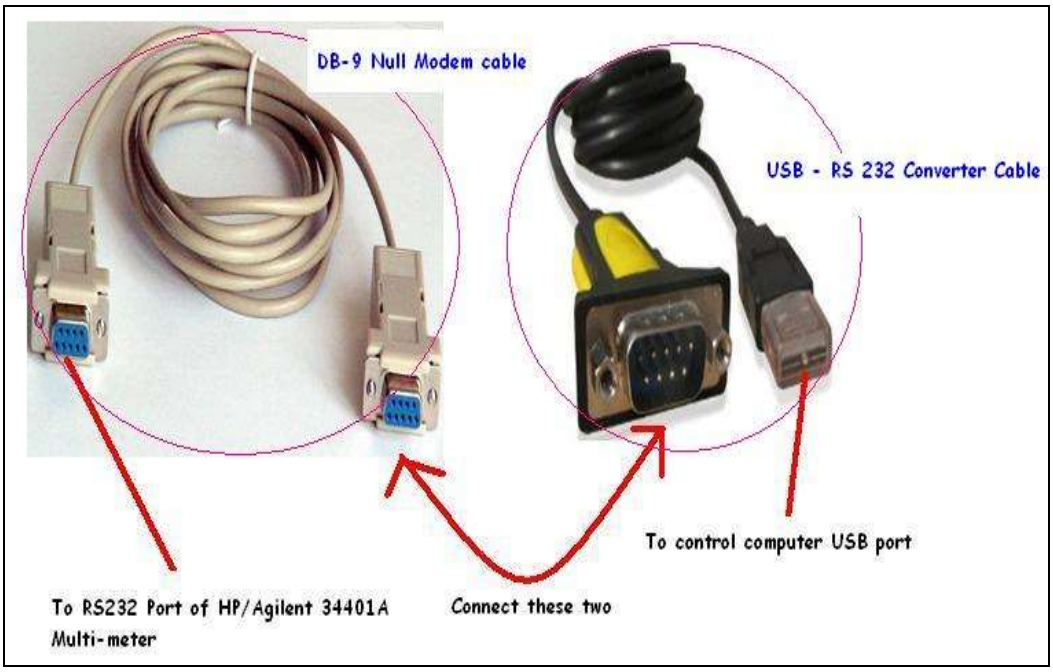


**Figure 3.2.1A** HP/Agilent 34401A Multi-meter Interface Cable

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### 3.2.2 DC Board Calibration Hardware Connections

1. Begin by removing the Output Adapter, if there is one, by gently pulling it off from the system test head. Output Adapter is used when set-up requires the use of Dual Die Mux (DDM).
2. Connect RS-232 to USB Cable between HP/Agilent Multi-Meter and control computer USB port.
3. Connect Ribbon Cable between CBits on Calibration Box and DC CBits on Test Head.
4. Connect DMM HI / LO from Calibration Box to Force HI / LO of HP/Agilent Multi-Meter.
5. Connect Sense HI / LO from Calibration Box to Sense Hi / LO of HP/Agilent Multi-Meter.
6. Connect Gate from Calibration Box to Gate Force on DC side of Test Head.
7. Connect Drain F from Calibration Box to Drain F on DC side of Test Head.
8. Connect Drain S from Calibration Box to Drain S on DC side of Test Head.
9. Connect Source F from Calibration Box to Source F on DC side of Test Head.
10. Connect Source S from Calibration Box to Source S on DC side of Test Head.

Refer to Figure 3.2.2A for the DC board calibration set-up.

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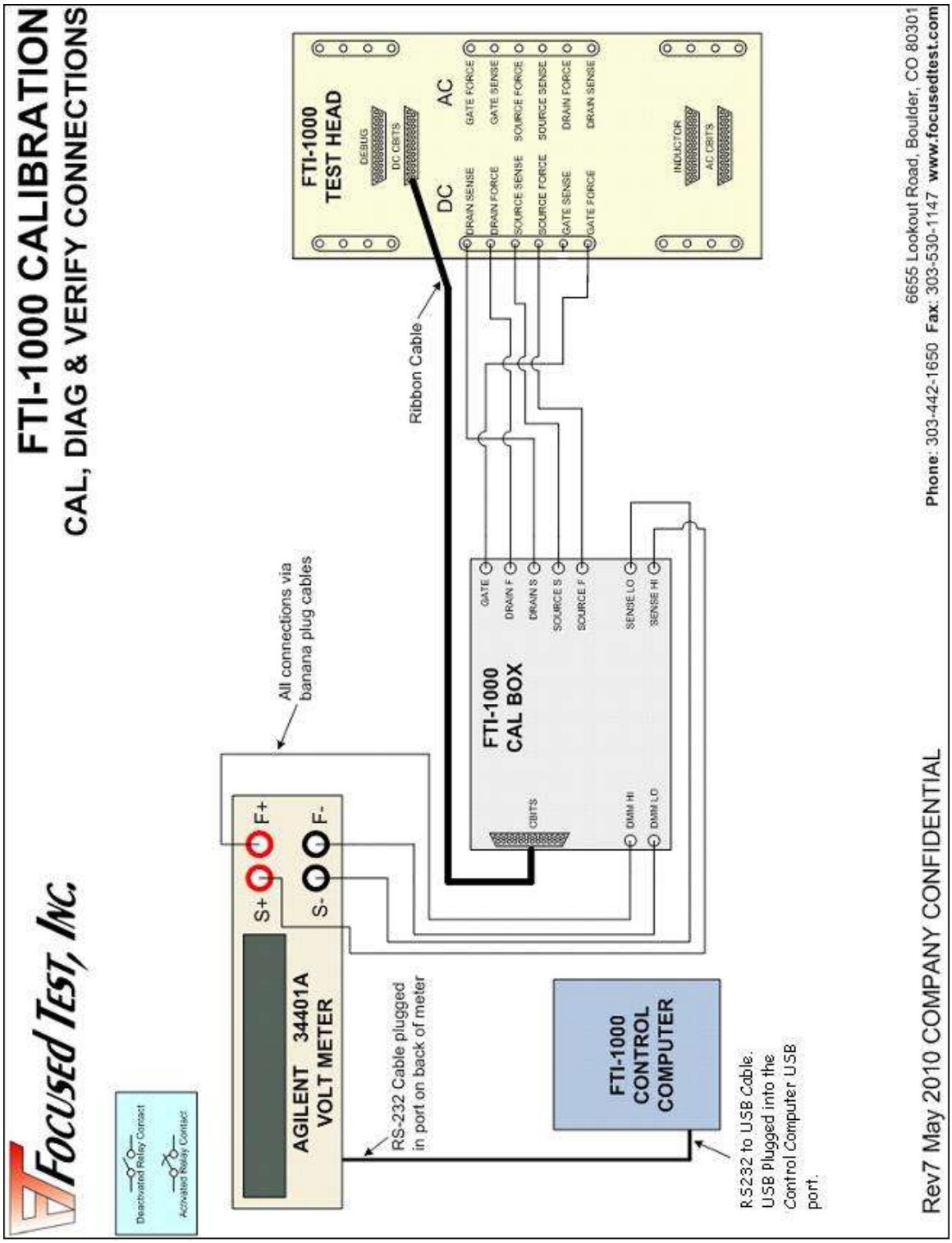


**Figure 3.2.2A** DC Board Calibration - Hardware Connections and Set-up

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### 3.2.3 AC Board Calibration Hardware Connections

1. Begin by removing the Output Adapter, if there is one, by gently pulling it off from the system test head. Output Adapter is used when set-up requires the use of Dual Die Mux (DDM).
2. Connect RS-232 to USB Cable between HP/Agilent Multi-Meter and control computer USB port.
3. Connect Ribbon Cable between CBits on Calibration Box and AC CBits on Test Head.
4. Connect DMM HI / LO from Calibration Box to Force HI / LO of HP/Agilent Multi-Meter.
5. Connect Sense HI / LO from Calibration Box to Sense Hi / LO of HP/Agilent Multi-Meter.
6. Connect Gate from Calibration Box to Gate Force on DC side of Test Head.
7. Connect Drain F from Calibration Box to Drain F on DC side of Test Head.
8. Connect Drain S from Calibration Box to Drain S on DC side of Test Head.
9. Connect Source F from Calibration Box to Source F on DC side of Test Head.
10. Connect Source S from Calibration Box to Source S on DC side of Test Head.
11. Short the User Gate connection found on the system Test Head.
12. Connect Inductor (of any value) at the User Inductor inputs found at the system Test Head.

Refer to Figure 3.2.3A for the AC board calibration set-up.

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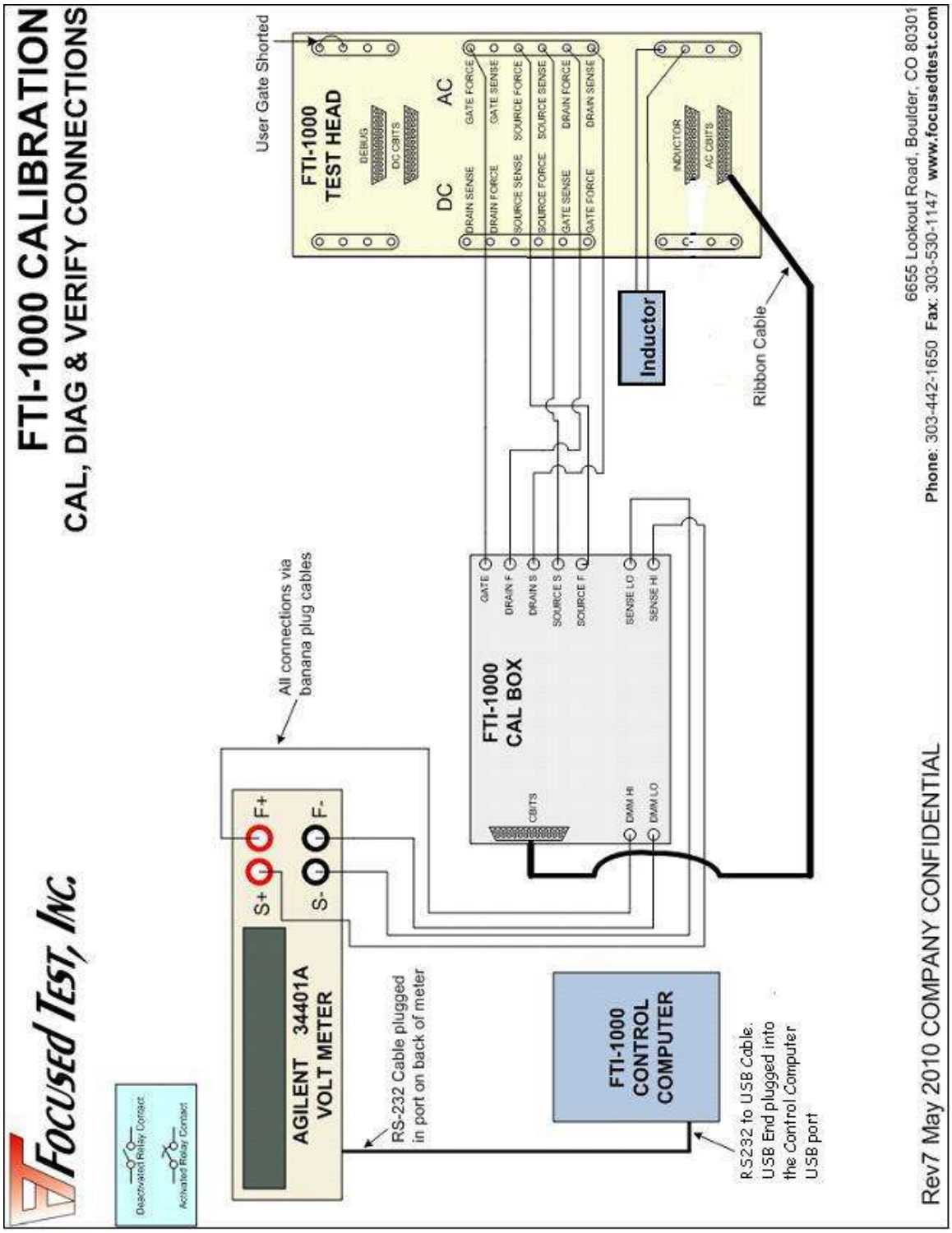


**Figure 3.2.3A** AC Board Calibration - Hardware Connections and Set-up

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## 3.3 Loading the Calibration Test Program

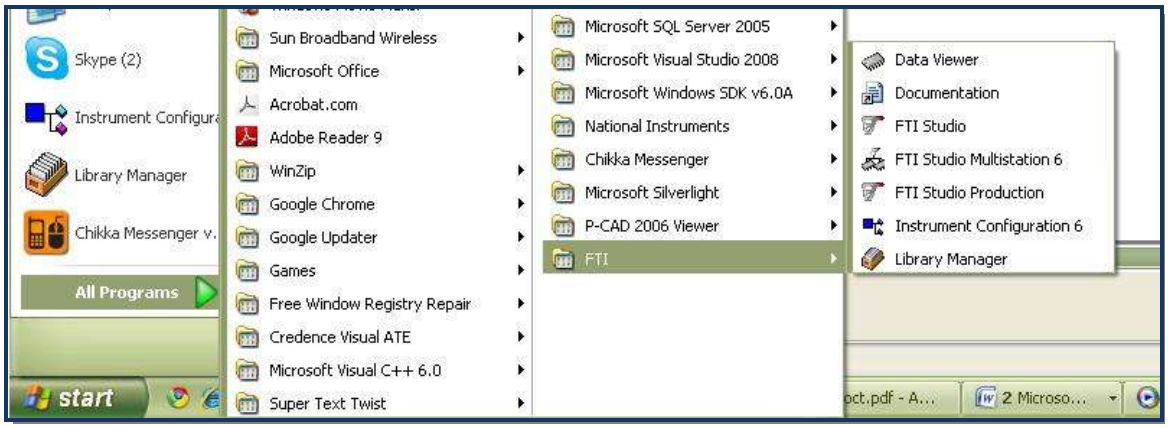
Follow steps to load Calibration Test Program.

1. Open FTI Studio.

There are two ways to open FTI Studio:

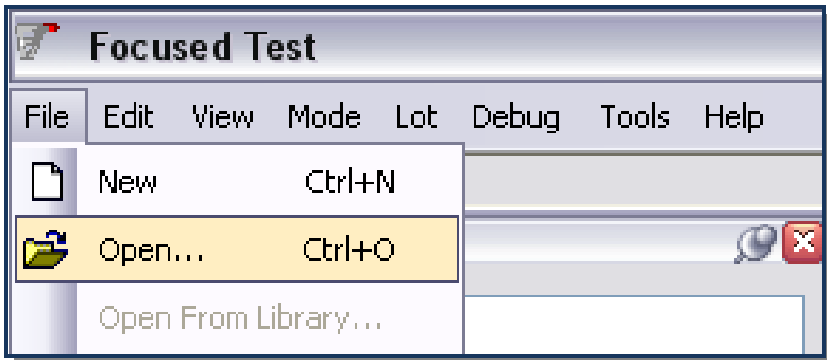
1A. Double click FTI Studio shortcut located on the desktop.

1B. Start Button → All Programs → FTI → FTI Studio




2. Open Calibration Test Program.

2A. Open the test program by clicking on the "File" button found on the menu tab. Then select "Open".

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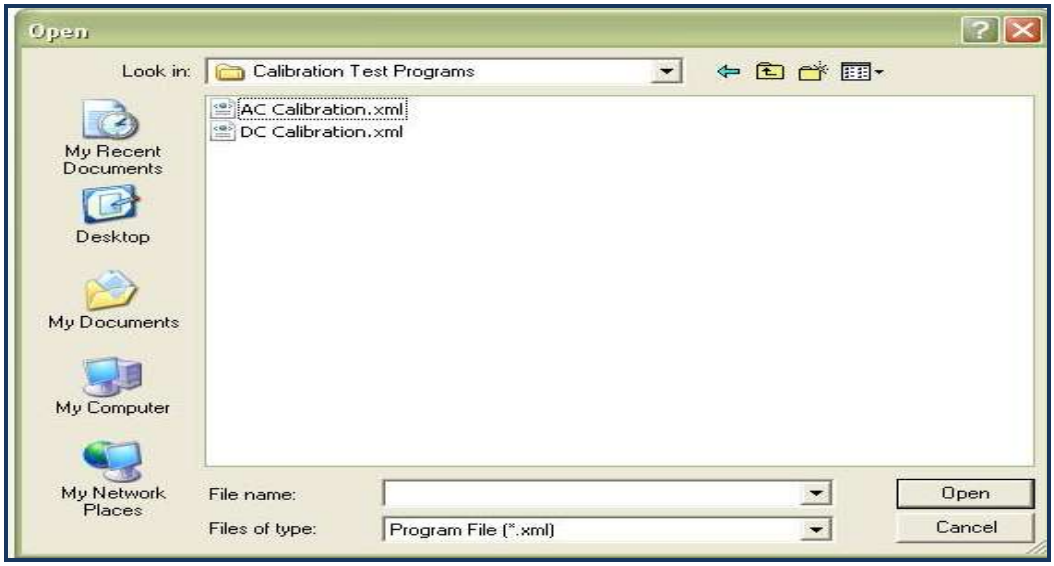
2B. Navigate to C:\Program Files\Focused Test Inc\Calibration Test Programs folder.

2B.1 DC Board Calibration select test program: **DC Calibration.xml**

2B.2 AC Board Calibration select test program: **AC Calibration.xml**




3. Test Program Overview.

After the calibration program opens, at the top of the screen there are a few pull down menus to make some selections with. This is where the different programs can be run, and choices for individual tests can be made.

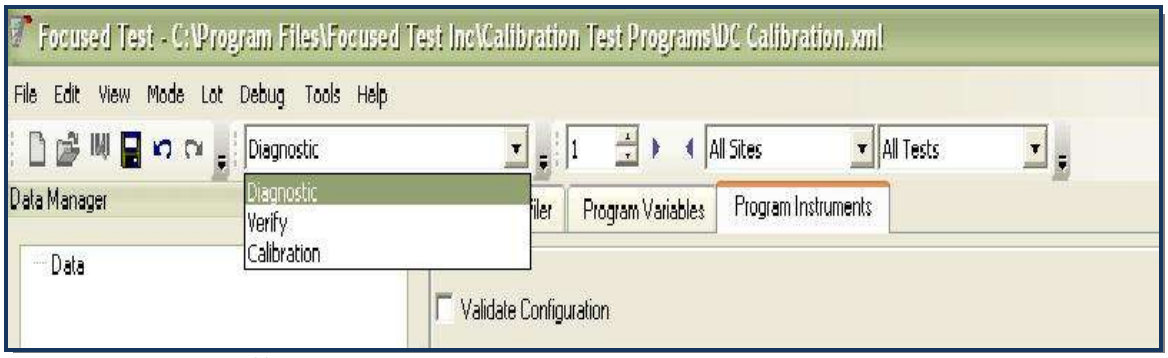



**Figure 3.3.3A** DC Calibration Test Program

- DC Diagnostics – Here there a different tests used for diagnosing the system to make sure the relays work and that the system is hooked up correctly on the DC motherboard. The subtests are typically run individually.
- DC Verify – This program runs tests that have settings made to only run the verification tests using the existing stored values for gains and offsets on the DC motherboard.

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- DC Calibration – This program runs tests that have settings made to calculate gains and offsets and store them in memory on the system and run verification using those gains and offsets on the DC motherboard.

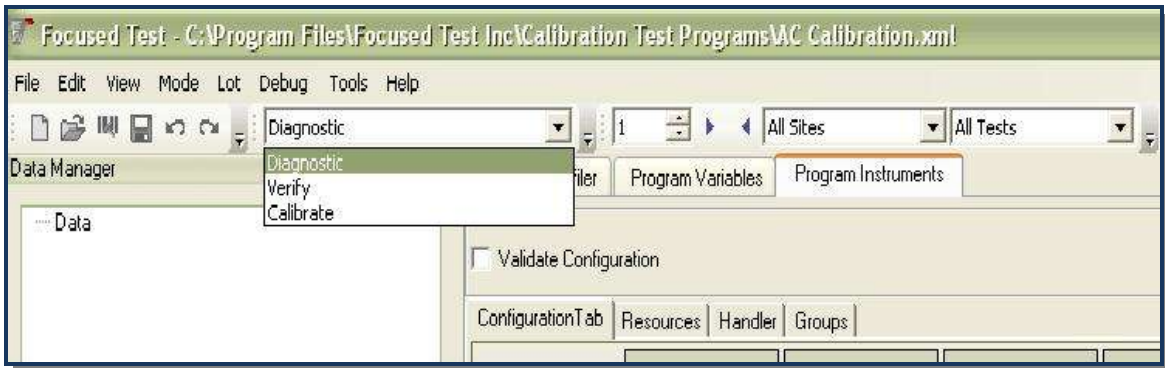


**Figure 3.3.3B** AC Calibration Test Program

- AC Diagnostics – Here there a different tests used for diagnosing the system to make sure the relays work and that the system is hooked up correctly on the AC motherboard. The subtests are typically run individually.

- AC Verify – This program runs tests that have settings made to only run the verification tests using the existing stored values for gains and offsets on the AC motherboard.

- AC Calibration – This program runs tests that have settings made to calculate gains and offsets and store them in memory on the system and run verification using those gains and offsets on the AC motherboard.

## 3.4 Running System Diagnostics, Verification, and Calibration

### 3.4.1 Running Diagnostic Routines

It is recommended that before going thru Verification and Calibration routines, Diagnostic routine should be run first. Diagnostics will ensure that relays on board are still functional and that specified calibration set-up or connections are done correctly.

To run Diagnostic tests, just select “Diagnostic” when you open the left side pull down menu (As shown in Figures 3.3.3A and 3.3.3B).

The pull down menu found on the right side will allow you to select the tests performed under Diagnostics.

It is recommended to run each test individually. Just select the test you want to run, and then click the play button. The play button will run the test you have selected.

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Refer to Figures 3.4.1A and 3.4.1B for DC and AC Diagnostic Tests.

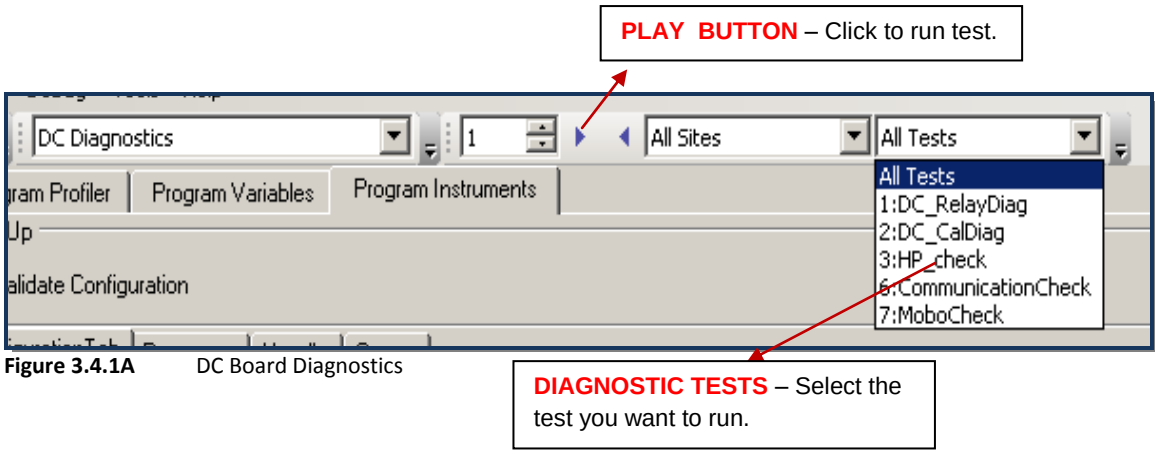


**Figure 3.4.1A** DC Board Diagnostics

- DC_RelayDiag – This test will check all the relays on the DC motherboard to ensure they are all operating correctly. It checks that they both open and close.
- DC_CalDiag – This test will check to make sure the Cal Board is connected and operating correctly. It also checks the resistors used for the calibration and verification tests. ***THIS TEST MUST BE PERFORMED WITH THE CALIBRATION BOX HOOKED UP FOR NORMAL CALIBRATION / VERIFICATION CONNECTIONS MADE.***
- HP_check – This test is used to check the High Power Module on the DC motherboard to make sure it is connected properly and that all the relays and analog switches are working correctly.
- CommunicationCheck – This test is used to check the USB communication to the DC Motherboard is functioning properly.
- MoboCheck – This test is used to check and verify the communication to the DC motherboard is functioning properly.

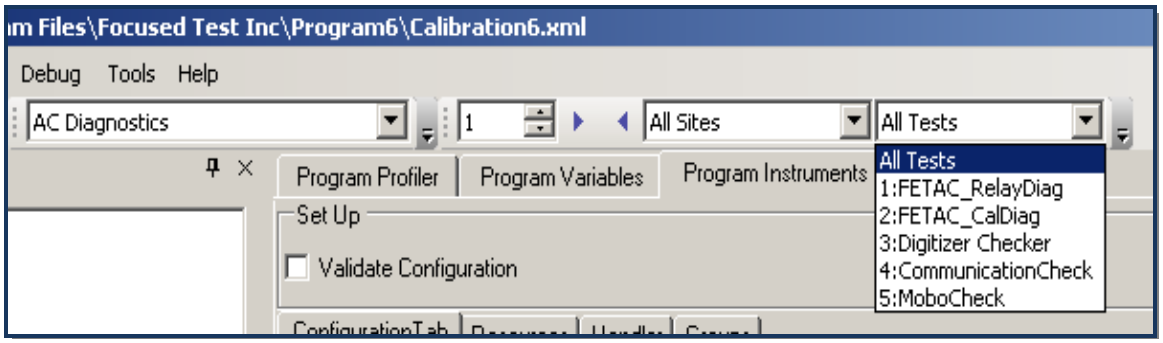


**Figure 3.4.1B** AC Board Diagnostics

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- FETAC_RelayDiag – This test will check all the relays on the AC Motherboard to ensure they are all operating correctly. It checks that they both open and close.
- FETAC_CalDiag – This test will check to make sure the Cal Board is connected and operating correctly. It also checks the resistors used for the calibration and verification tests. ***THIS TEST MUST BE PERFORMED WITH THE CALIBRATION BOX HOOKED UP FOR NORMAL CALIBRATION / VERIFICATION CONNECTIONS MADE.***
- Digitizer Checker – This test is used to check the Digitizer Module on the AC motherboard to make sure it is connected properly and that all the relays and analog switches are working correctly.
- CommunicationCheck – This test is used to check the USB communication to the AC Motherboard is functioning properly.
- MoboCheck – This test is used to check and verify the communication to the AC motherboard is functioning properly.

### 3.4.2 Running Verification Routine

Verification can be run to check the system to ensure it meets the specifications. This routine is performed after the Diagnostics.

Running the Verification routine is similar with the procedure performed during the Diagnostics.

1. Select "Verification" when you open the left side pull down menu.
2. Open the pull down menu found at the right side to select the Verification tests you want to run. It is recommended to run each test individually and in order as some tests depend on the results from previous tests to run correctly. Refer to Figure 3.4.2A and Figure 3.4.2B for the DC and AC Verification tests.
3. Run test by clicking the "Play" button.

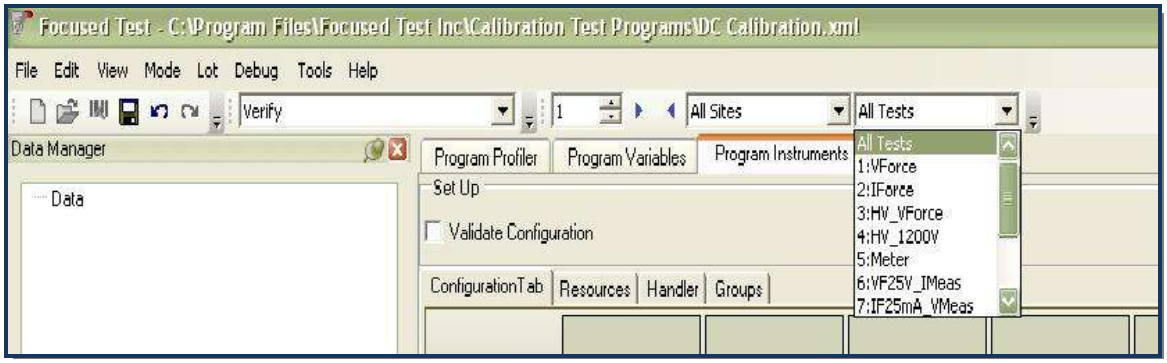


**Figure 3.4.2A** DC Board Verification

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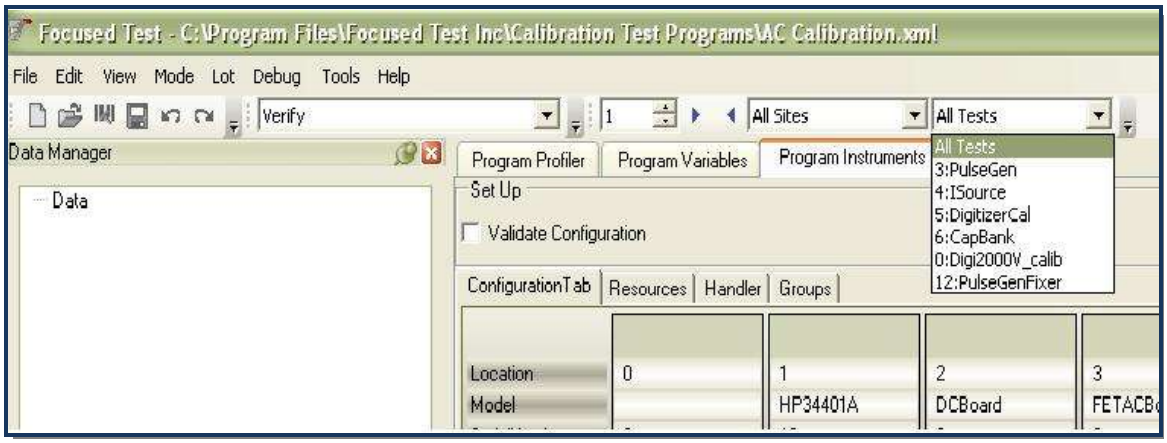


**Figure 3.4.2B** AC Board Verification

### 3.4.3 Running Calibration Routine

Calibration routine are often performed when test/s fail during the Verification routine. When a calibration is run Gains and Offsets are calculated and stored and then Verification is performed afterwards as well.

Calibration typically takes about 15 minutes to perform.

Calibration routine is performed similar to the Diagnostics and Verification routine procedures. It is recommended that the tests will be run individually or one after the other and in order as some tests depend on the results of the previous tests run.

1. Select “Calibration” when you open the left side pull down menu.

2. Open the pull down menu found at the right side to select the Calibration tests you want to run. It is recommended to run each test individually and in order as some tests depend on the results from previous tests to run correctly. Refer to Figure 3.4.3A and Figure 3.4.3B for the DC and AC Calibration tests.

3. Run test by clicking the “Play” button.

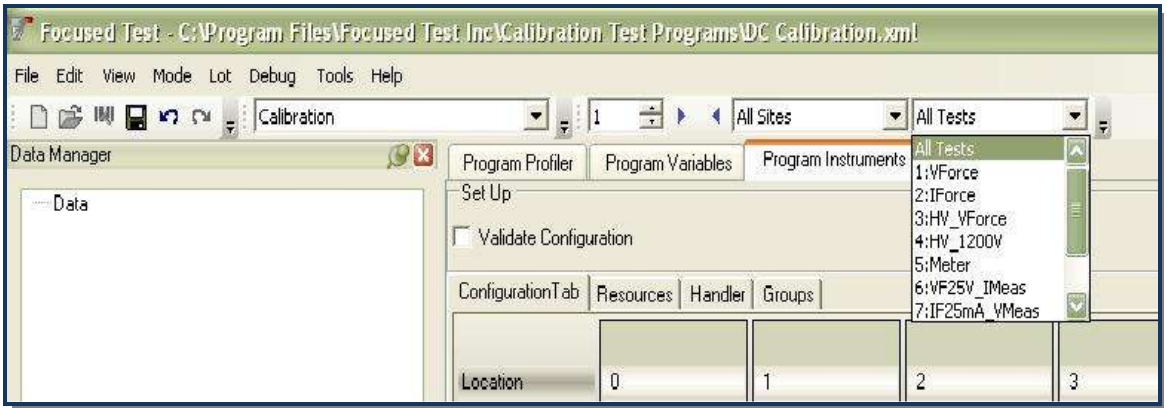


**Figure 3.4.3A** DC Board Calibration

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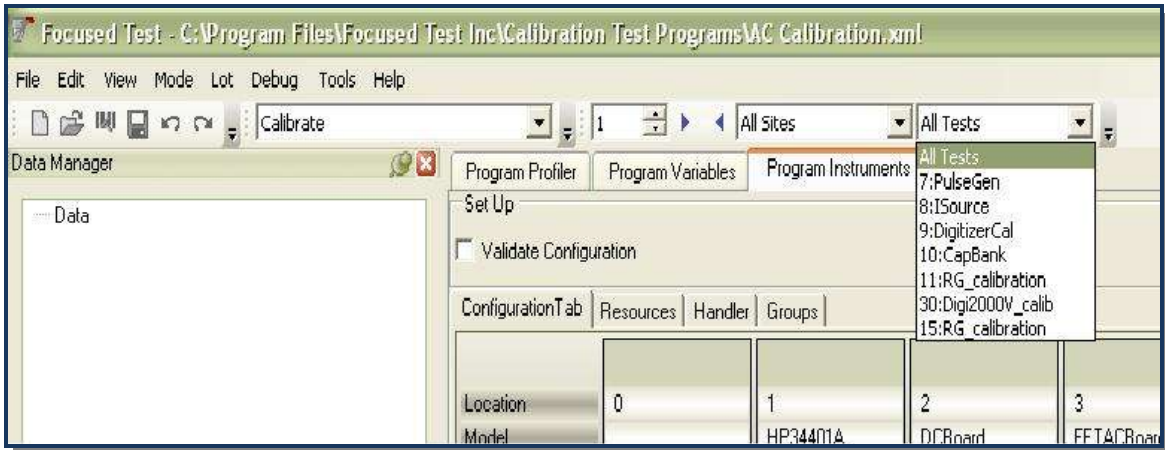


**Figure 3.4.3B** AC Board Calibration

## 3.5 Printing Results

After running either the Calibration or Verification routine the data logs generated may be printed out for a hard copy. Data logs may also be saved out for storage on a server for later review.

## 3.6 Temperature Stability

FTI guarantees its systems will remain in specification after calibration for +/- 3deg F for a period of 1 month.

## 3.7 Calibration Intervals

The standard FTI calibration interval is 1 month. However, calibration intervals are a risk tradeoff if shipped product is subject to recall. Running verification on a shorter interval than the standard calibration interval can reduce exposure to recall.

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